

All Commands Used — OS Lab Reports 2 to 6

ICT-3102: Operating System Lab | Md. Shaon Khan | ID: 1984

Lab Report 2 — Linux File Commands (File & Directory Management)

#	Command	Description
1	ls	Lists all files and directories in the current folder
2	ls -al	Detailed list including hidden files
3	ls -lt	Lists files sorted by modification time, newest first
4	cd <dir>	Changes to the specified directory (e.g., cd Documents)
5	cd -	Goes back to the previously visited directory
6	cd ..	Moves up one level to the parent directory
7	cd	Goes directly to the home directory
8	pwd	Prints the full path of the current working directory
9	mkdir <dir>	Creates a new directory (e.g., mkdir Projects)
10	cat > <file>	Creates a new file and writes typed input into it (e.g., cat > notes.txt)
11	echo "text" >> <file>	Appends a line of text to an existing file
12	more <file>	Shows file content page by page
13	head <file>	Displays the first 10 lines of a file
14	tail <file>	Displays the last 10 lines of a file
15	touch <file>	Creates an empty file or updates its timestamp
16	rm <file>	Deletes the specified file (e.g., rm hello.txt)
17	rm -r <dir>	Deletes a directory and everything inside it (e.g., rm -r Projects)
18	rm -f <file>	Force deletes a file without confirmation
19	rm -rf <dir>	Force deletes a directory and all contents recursively
20	clear	Clears the terminal screen
21	echo -e	Interprets escape characters like \n (used when adding numbered lines)

Lab Report 3 — Copy, Move, Rename, grep, Vim

#	Command	Description
1	cp file1 file2	Copies file1 to file2
2	cat file1.txt >> file2.txt	Appends content of file1 to file2
3	cp -r dir1 dir2	Recursively copies a directory (e.g., cp -r Projects folder1)
4	mv file1 file2	Renames/moves a file (e.g., mv old_name.txt new_name.txt)
5	mv file /path/dir/	Moves a file to another directory
6	cp file /path/dir/	Copies a file to another directory
7	cp file .	Copies a file to the current directory
8	grep "pattern" file	Searches for a pattern in a file
9	grep -i "pattern" file	Case-insensitive pattern search
10	grep -n "pattern" file	Shows matching lines with line numbers
11	grep -r "pattern" dir/	Recursively searches all files in a directory (e.g., grep -r "main()" codes/)
12	grep -v "pattern" file	Shows lines NOT containing the pattern
13	vim filename.txt	Opens a file in the Vim text editor
14	sudo apt install vim -y	Installs Vim editor
15	su -	Switches to root user
16	usermod -aG sudo shaon	Adds user shaon to the sudo group

Vim keys used: i (insert mode), Esc (return to normal mode), :w (save), :q (quit), :wq (save & quit), :q! (quit without saving)

Lab Report 4 — Unix File Permission / Access Mode

#	Command	Description
1	ls	Lists files and directories
2	ls -l	Detailed listing with permissions, owner, size, timestamps
3	ls -l hello.txt	Detailed listing for a specific file
4	ls file1*.txt	Wildcard listing — files starting with “file1”, ending “.txt”
5	ls fi*.txt	Wildcard listing — files starting with “fi”
6	ls *.txt	Lists all .txt files
7	mkdir OS Lab	Creates two new directories, OS and Lab
8	cd OS	Changes into the OS directory
9	touch file1.txt file12.txt file2.txt notes.txt file1.pdf	Creates multiple empty files at once
10	cat > notes.txt	Creates a file and writes content into it
11	cat > routine.txt	Creates another file with typed content
12	cat hello.txt	Displays file contents
13	cat -b hello.txt	Displays contents with line numbers for non-blank lines

14	<code>wc hello.txt</code>	Counts lines, words, characters in a file
15	<code>more Hello1.txt</code>	Displays file content one screen at a time
16	<code>vi hello.txt</code>	Opens the vi editor to create/edit a file
17	<code>chmod g+x hello.txt</code>	Adds execute permission for group
18	<code>chmod o+wx hello.txt</code>	Adds write+execute permission for others
19	<code>chmod u+x hello.txt</code>	Adds execute permission for owner
20	<code>chmod u-rwx hello.txt</code>	Removes read/write/execute for owner
21	<code>chmod g-x hello.txt</code>	Removes execute for group
22	<code>chmod o-r hello.txt</code>	Removes read for others
23	<code>chmod o=x hello.txt</code>	Sets others' permission to execute only
24	<code>chmod g=x hello.txt</code>	Sets group's permission to execute only
25	<code>chmod u=r hello.txt</code>	Sets owner's permission to read only
26	<code>chmod 444 hello.txt</code>	Numeric mode — read-only for all
27	<code>chmod 000 hello.txt</code>	Numeric mode — removes all permissions
28	<code>chmod 777 hello.txt</code>	Numeric mode — full permission for all
29	<code>chmod 562 hello.txt</code>	Numeric mode — r-x / rw- / -w-
30	<code>cd</code>	Returns to home directory

vi keys used: i (insert mode), Esc, :wq (save & quit)

Lab Report 5 — Users, Groups, and File Permissions

#	Command	Description
1	<code>getent passwd</code>	Displays all user accounts from /etc/passwd
2	<code>compgen -u</code>	Lists usernames only (shell built-in)
3	<code>sudo adduser <user></code>	Creates a new user account (e.g., <code>sudo adduser sehrish</code>)
4	<code>sudo passwd <user></code>	Sets/resets a user's password (e.g., <code>sudo passwd sehrish</code>)
5	<code>su <user></code>	Switches user without loading login environment (e.g., <code>su sehrish</code>)
6	<code>su - <user></code>	Switches user with full login environment (e.g., <code>su - sehrish</code>)
7	<code>ls</code>	Lists directory contents
8	<code>cd</code>	Changes directory
9	<code>pwd</code>	Prints working directory
10	<code>mkdir Downloads</code>	Creates the Downloads directory
11	<code>logout</code>	Exits current user session back to previous user
12	<code>ls /home</code>	Lists home directories

13	<code>sudo userdel -r <user></code>	Deletes a user and their home directory (e.g., <code>sudo userdel -r sehrish</code>)
14	<code>sudo groupadd <group></code>	Creates a new group (e.g., <code>sudo groupadd phyra</code>)
15	<code>sudo tail /etc/group</code>	Shows the last 10 lines of <code>/etc/group</code> to verify new groups
16	<code>groups / groups [user]</code>	Lists group memberships
17	<code>sudo usermod -aG <groups> <user></code>	Adds a user to supplementary groups
18	<code>id <user></code>	Shows UID, GID, and group memberships (e.g., <code>id shayankhan</code>)
19	<code>touch testfile.txt</code>	Creates a test file
20	<code>ls -l testfile.txt</code>	Shows detailed listing/permissions of a file
21	<code>sudo chown <user> <file></code>	Changes file owner (e.g., <code>sudo chown shayankhan testfile.txt</code>)
22	<code>chgrp <group> <file></code>	Changes file group ownership (fails without sudo, as shown)
23	<code>sudo chgrp <group> <file></code>	Changes file group ownership with elevated privileges
24	<code>chmod ug+s Downloads</code>	Applies SUID + SGID bits to a directory

Lab Report 6 — Introduction to Shell Scripting

#	Command / Construct	Description
1	<code>#!/bin/bash</code>	Shebang line specifying the Bash interpreter
2	<code>vim</code>	Text editor used to create/edit shell scripts
3	<code>more</code>	Displays file content one screen at a time
4	<code>chmod +x</code>	Grants execute permission to a script
5	<code>chmod -x</code>	Removes execute permission from a script
6	<code>./script.sh</code>	Executes a script from the current directory
7	<code>sh script.sh</code>	Runs a script via the sh interpreter (no execute bit required)
8	<code>ls / ls -l</code>	Lists directory contents / with permission details
9	<code>echo</code>	Prints text or variable values
10	<code>echo -e</code>	Interprets escape sequences (<code>\n</code> , <code>\t</code>)
11	<code>echo -n</code>	Prints text without a trailing newline
12	<code>printf</code>	Formats and prints data using C-style specifiers
13	<code>#</code>	Comment line (ignored during execution)
14	<code>read</code>	Reads user input from stdin into a variable

15	<code>\$VARIABLE</code>	References a variable's value
16	<code>\$HOME</code>	Environment variable — user's home directory path
17	<code>\$SHELL</code>	Environment variable — path of the active shell
18	<code>\$(command)</code>	Command substitution (e.g., <code>\$(date)</code>)
19	<code>\$(())</code>	Arithmetic expansion for integer math
20	<code>if ... fi</code>	Conditional block
21	<code>if ... else ... fi</code>	Two-branch conditional
22	<code>if ... elif ... else ... fi</code>	Multi-branch conditional
23	<code>case ... esac</code>	Pattern-matching multi-branch statement
24	<code>-eq / -ne</code>	Numeric equal to / not equal to
25	<code>-gt / -ge</code>	Numeric greater than / greater than or equal to
26	<code>-lt / -le</code>	Numeric less than / less than or equal to
27	<code>for i in list</code>	Iterates over a list of values
28	<code>for ((; ;))</code>	C-style numeric for loop
29	<code>while ... do ... done</code>	Loops while condition is true
30	<code>until ... do ... done</code>	Loops until condition becomes true
31	<code>break</code>	Exits the enclosing loop
32	<code>continue</code>	Skips to the next loop iteration
33	<code>+ - * / %</code>	Arithmetic: add, subtract, multiply, divide, modulo
34	<code>**</code>	Arithmetic: exponentiation
35	<code>function_name()</code>	Defines a reusable named function
36	<code>\$1, \$2</code>	Positional parameters (function/script arguments)
37	<code>return</code>	Exits a function, sets its exit status code
38	<code>\$?</code>	Special variable — exit status of the last command/function

Scripts created in this lab (all via vim → chmod +x → ./script.sh):

welcome.sh, variable.sh, comment.sh, env.sh, date.sh, input.sh, if.sh, iselse.sh, grade.sh, choice.sh, for.sh, cfor.sh, while.sh, unt.sh, breakcon.sh, arith.sh, function.sh, return.sh